Byzantine Reliable Broadcast

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Byzantine reliable broadcast

- Almost the same primitive as the fail-silent model. It needs to reach agreement on delivered messages.
- To simplify the protocol, a Byzantine reliable broadcast instance is used to deliver **one message**.
- A priori declares a **sender process** for the instance

Authenticated communication primitives

- Recall modules in model with crash failures
  - Perfect Links (pl)
  - Best-effort Broadcast (beb)
- Authenticated versions can be defined that tolerate attacks
  - Authenticated Perfect Links (al)
  - Authenticated Best-effort Broadcast (abeb)
- Implemented using cryptographic authentication (MACs or digital signatures)

Byzantine broadcast variants

- Byzantine consistent broadcast
- Byzantine reliable broadcast

Byzantine Consistent Broadcast (bcb)

Events

- Request <broadcast (m)>
  Broadcasts a message m to all processes
- Indication <deliver (p, m)>
  Delivers a message m from sender p

Byzantine Consistent Broadcast (bcb)

Properties:

- BCB1 (Validity) = BEB1:

  Every message broadcast by a correct process is eventually delivered by every correct process.
- BCB2 (No duplication):

  Every correct process delivers at most one message.
- BCB3 (Integrity):

  If a correct process delivers m with sender p, and p is correct, then p has broadcast m.
- BCB4 (**Consistency**):

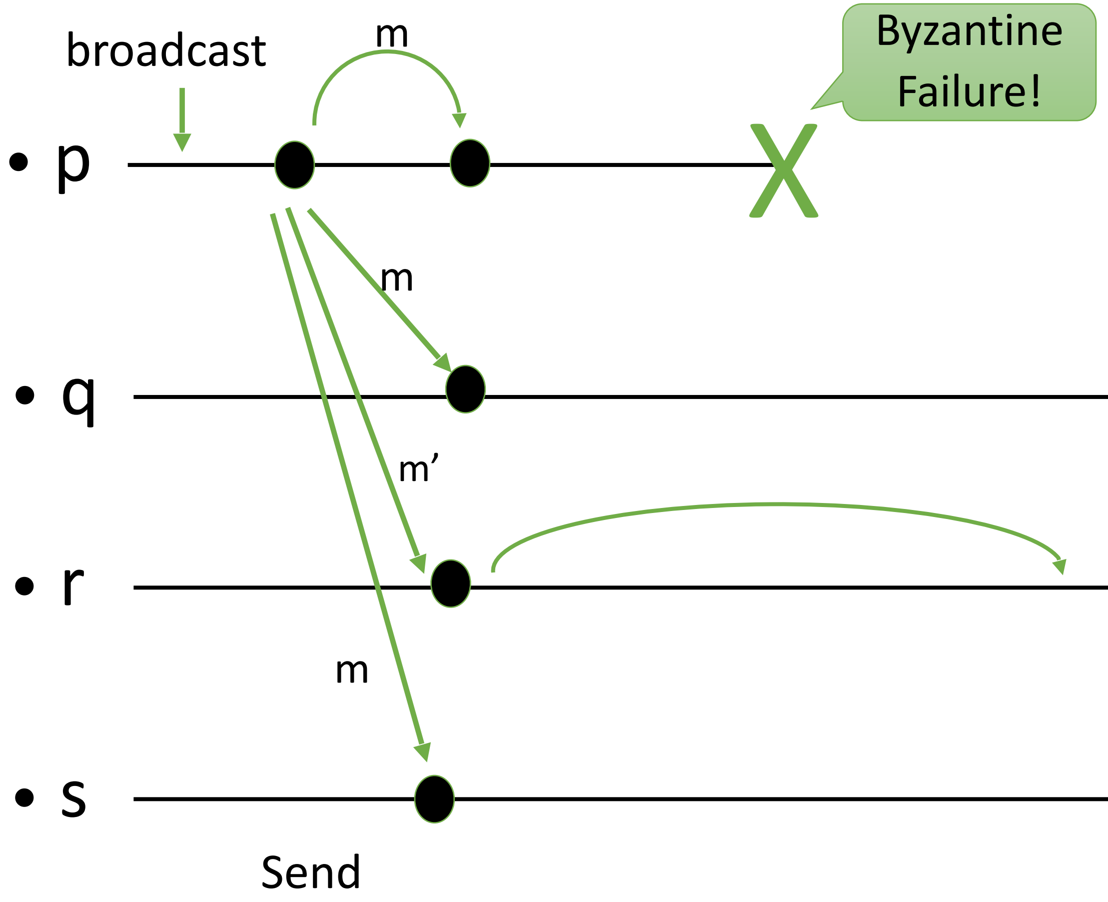
  If a correct process delivers message m and another correct process delivers message m', then m=m'.

- Note: Some correct process may not deliver any message (agreement is not required yet)

Using Byzantine Quorums

- System of $N > 3f$ processes, f are faulty.
- Let's have $N = 3f + 1$ for simplicity.
- Every subset with size larger than or equal to $2f+1$ processes is a quorum.
- The collection of all quorums is a quorum system.

Example



- Faulty sender p sends two different message m and m'.
- Processes q and s deliver the message m.

- Process r does not deliver any message. It does not receive the same message from a quorum.
- 2 rounds, $O(n^2)$ messages
  $O(n^2\ |m|)$ communicated data

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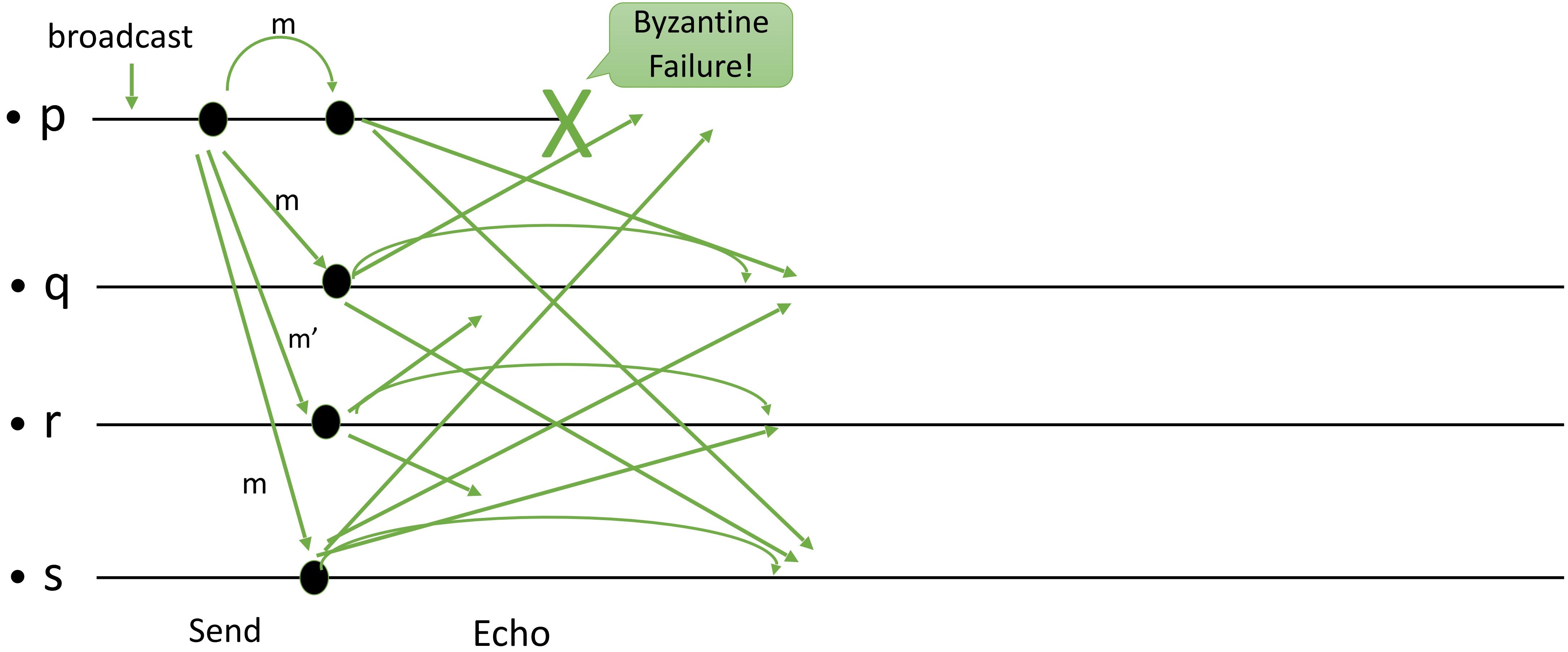

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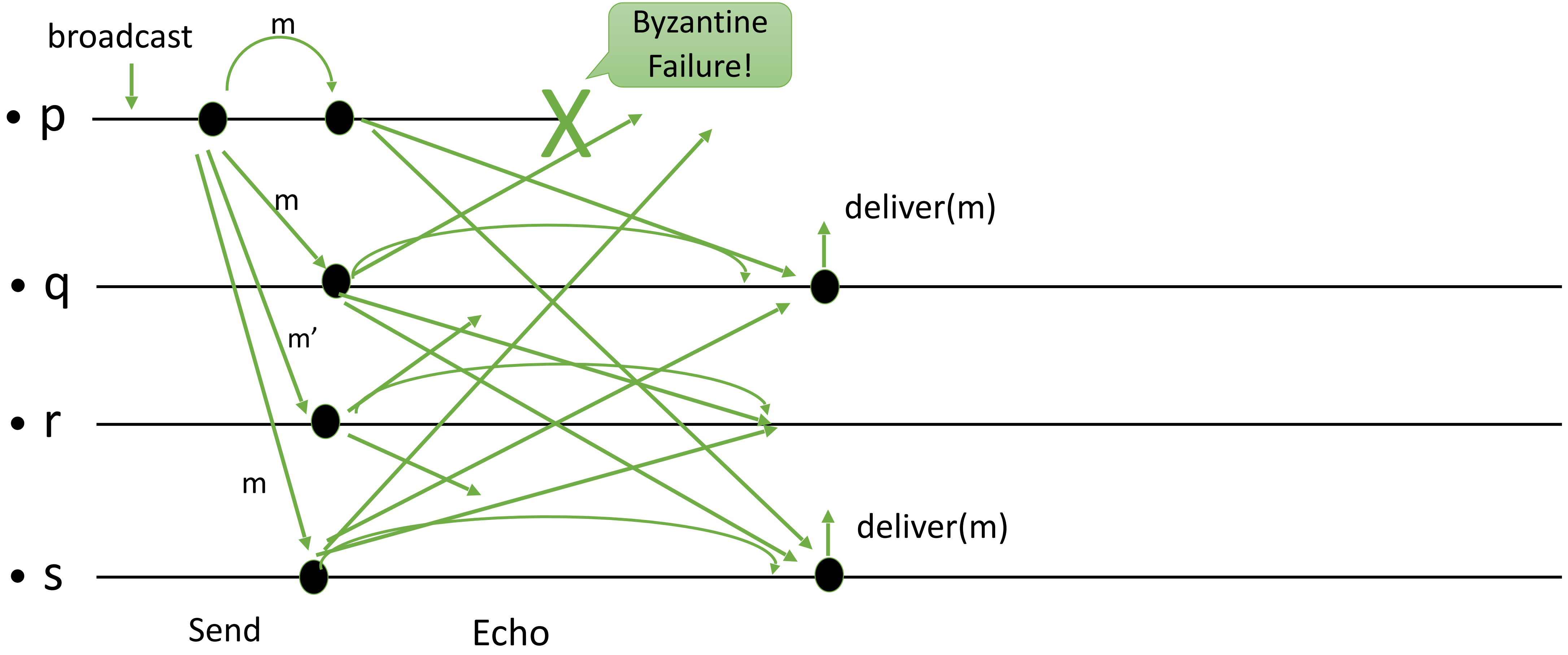

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Authenticated Echo Broadcast

Implements bcb, **uses** abeb, with sender s (where $N = 3f + 1$)

upon <bcb, broadcast (m)> **do**

    trigger <abeb, broadcast(Send(m))>

upon <abeb, deliver(s, Send(m))> **do**

    trigger <abeb, broadcast(Echo(m))>

upon <abeb, deliver(p, Echo(m))> **do**

    echo[p] := m

    if $\exists m : \#\{p \mid echo[p]=m\} \geq 2f + 1$ **then**

        trigger <deliver(s, m)>

Note: The code to prevent duplicate execution is omitted.

Using Byzantine Quorums

Proof of the consistency property:

- Two distinct quorums together contain more than 4f+2 processes.
- Thus, they overlap in at least f + 1 processes.
- At least one of them is a correct process.
- The correct process in the intersection has beb broadcast the same message Echo[m] to all processes.
- Therefore, the two processes deliver the same message m.

Byzantine Reliable Broadcast (brb)

Events

- Request <broadcast(m)>
- Indication <deliver(p, m)>

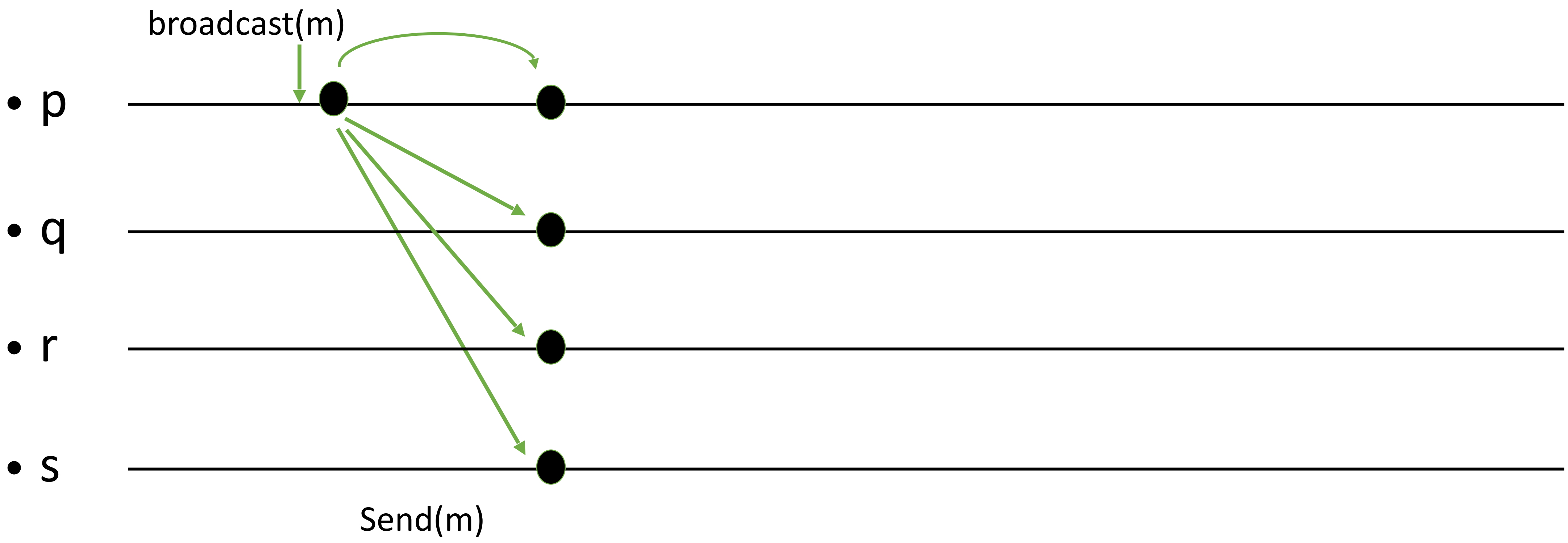
Properties

- BRB1 (Validity) = BCB1
- BRB2 (No duplication) = BCB2
- BRB3 (Integrity) = BCB3
- BRB4 (Consistency) = BCB4
- BRB5 (**Totality**):

  If some correct process delivers a message, then every correct process eventually delivers a message.

By totality, either all or none of the correct processes deliver a message. By consistency, they deliver the same message.

Example, all correct



3 rounds; $O(n^2)$ messages; $O(n^2\ |m|)$ communication

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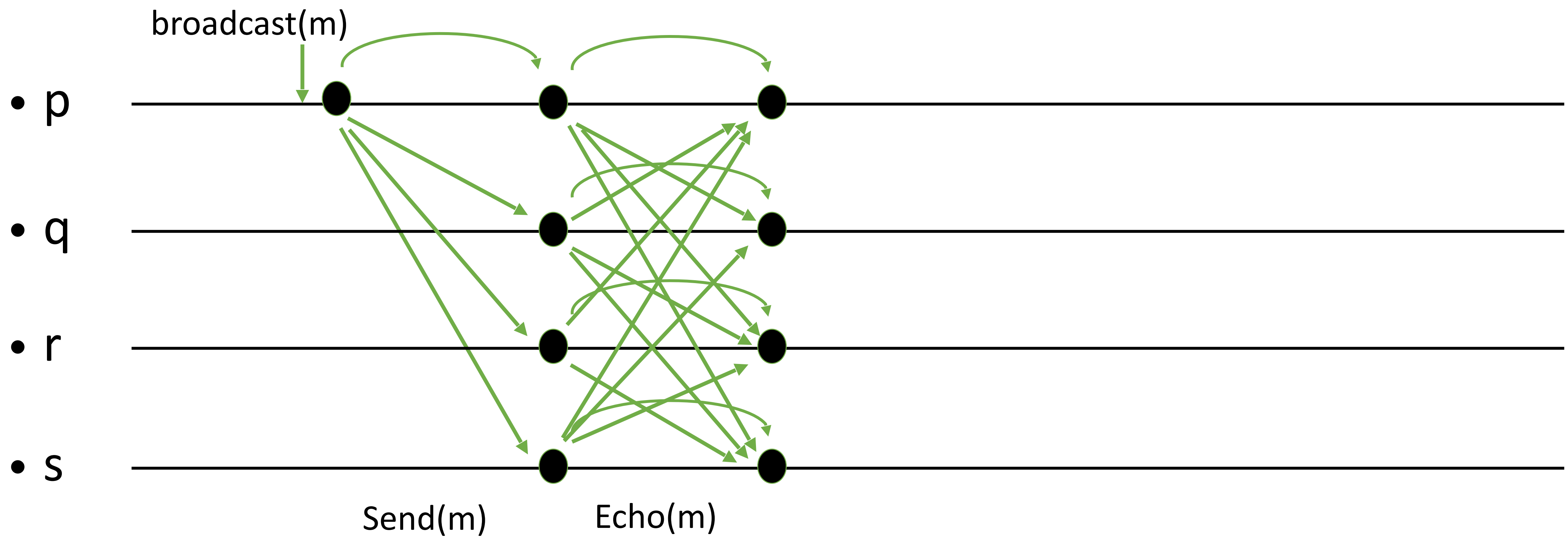

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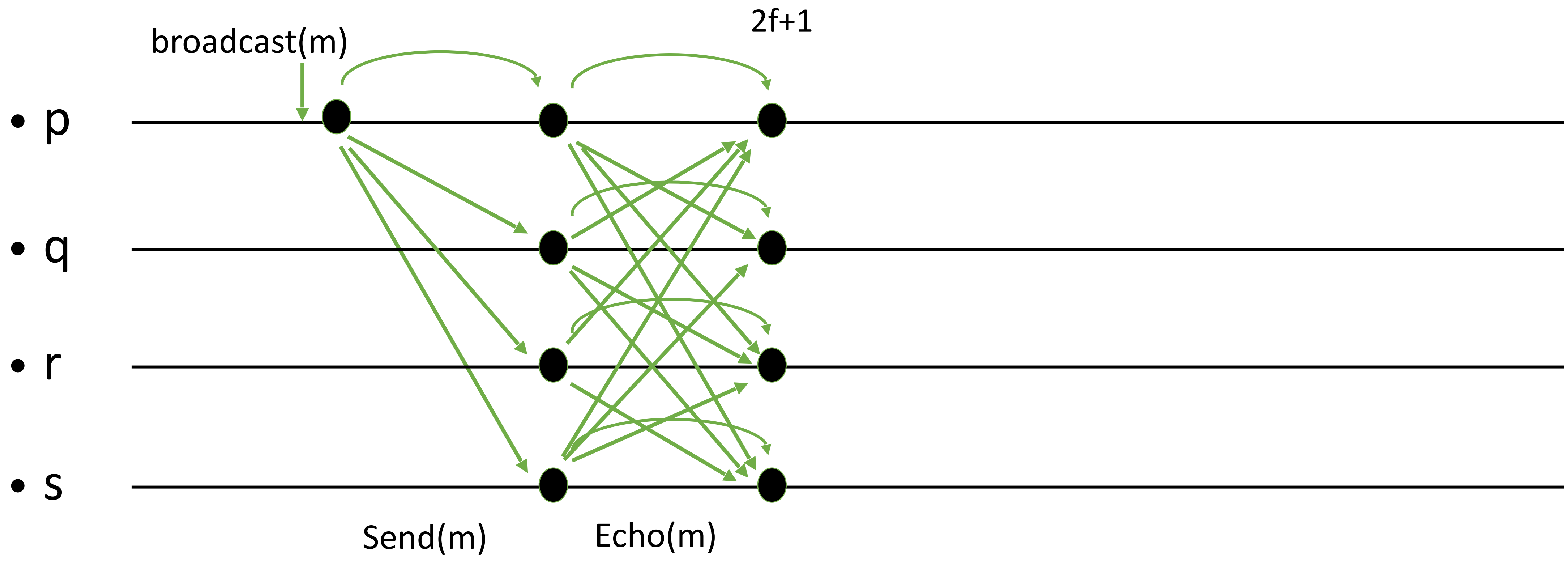

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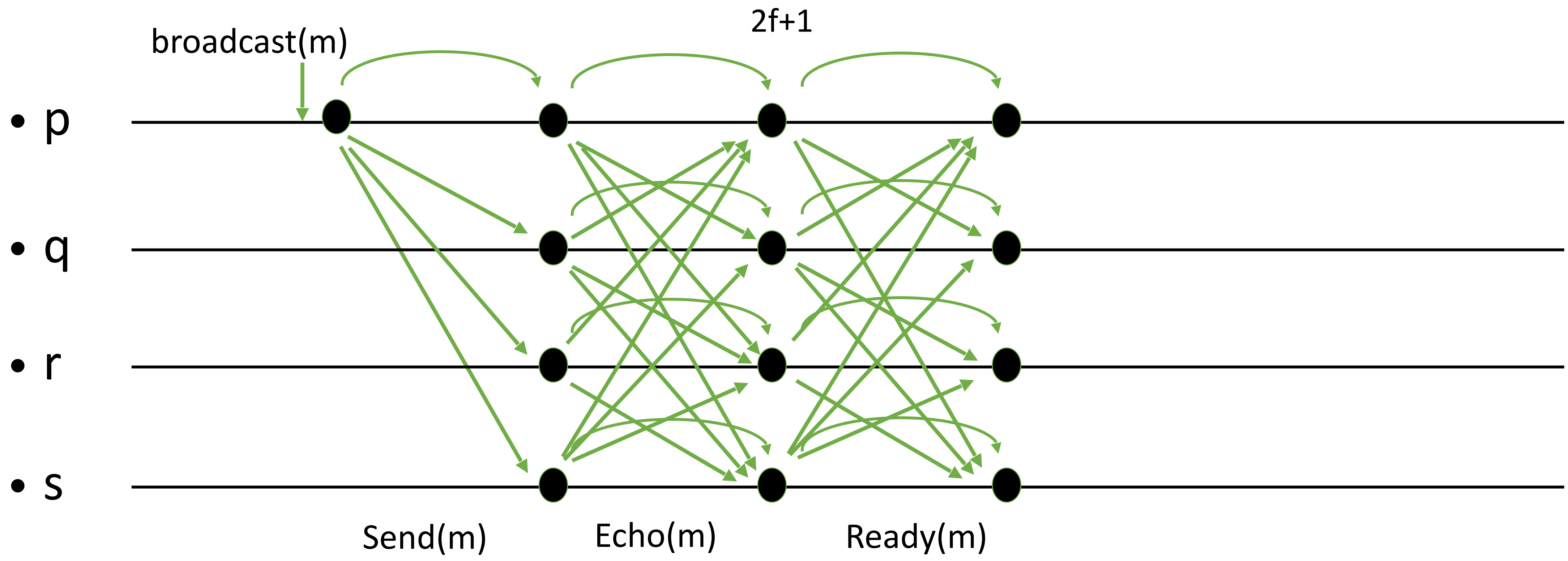

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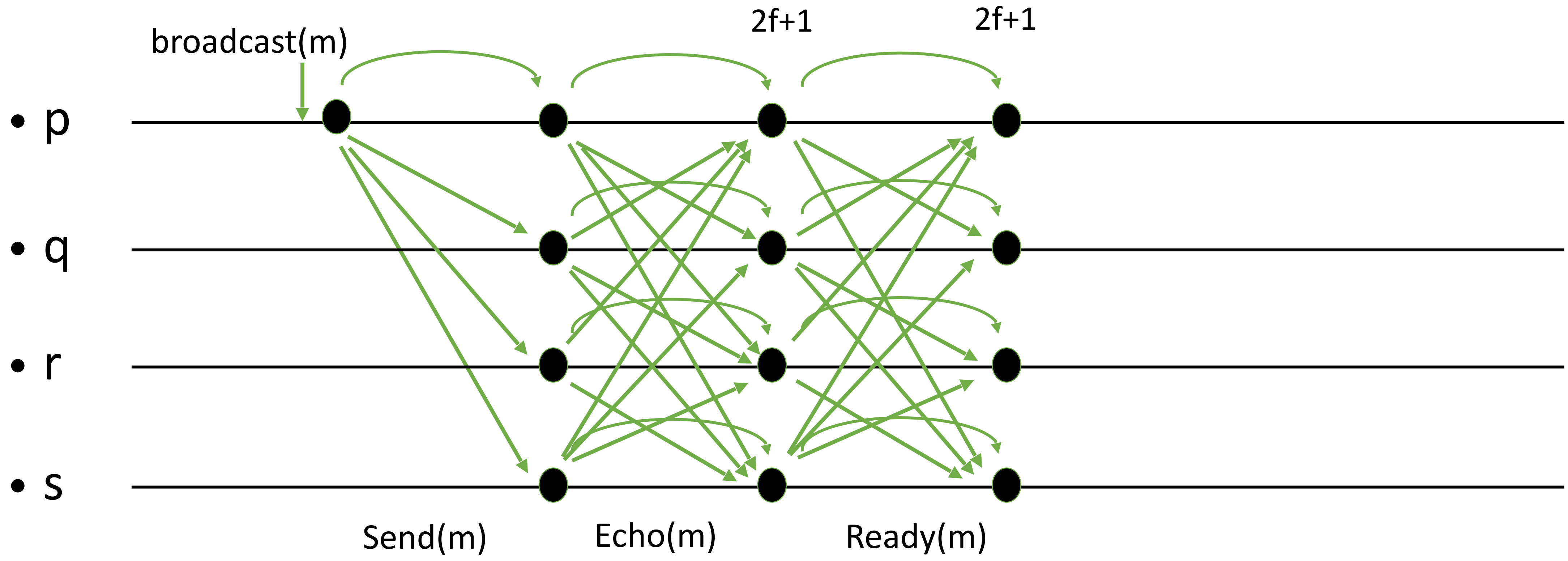

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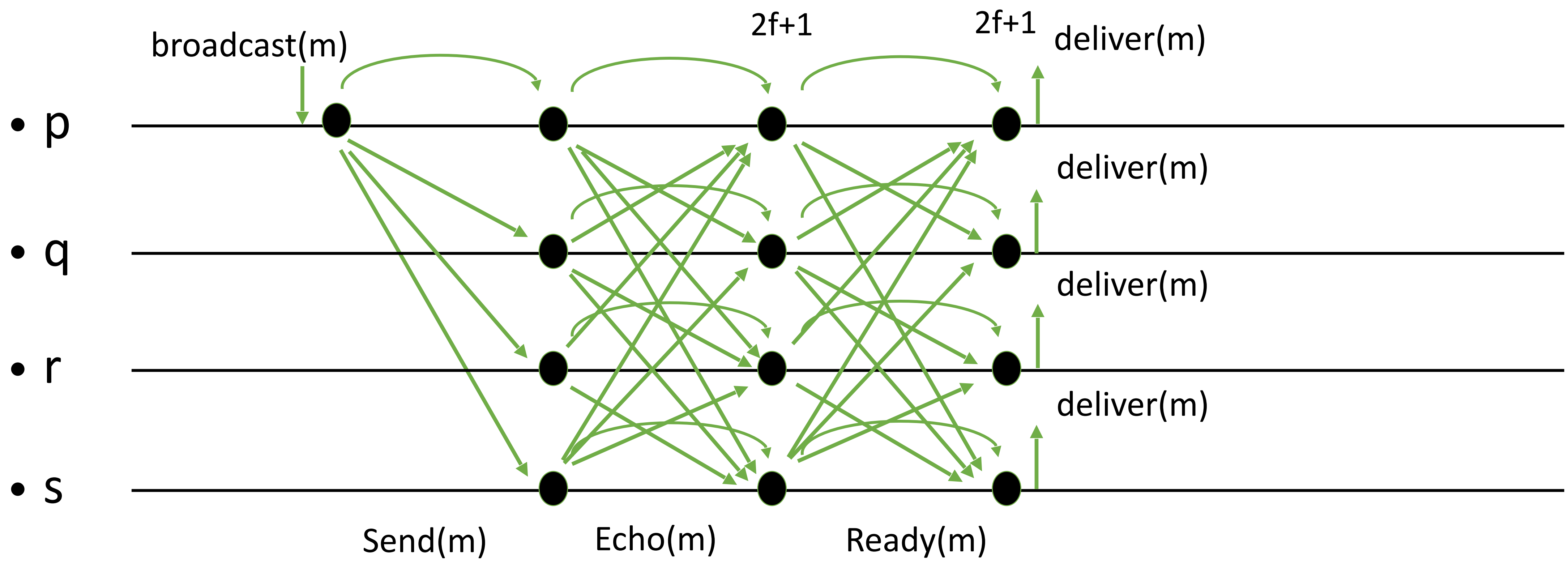

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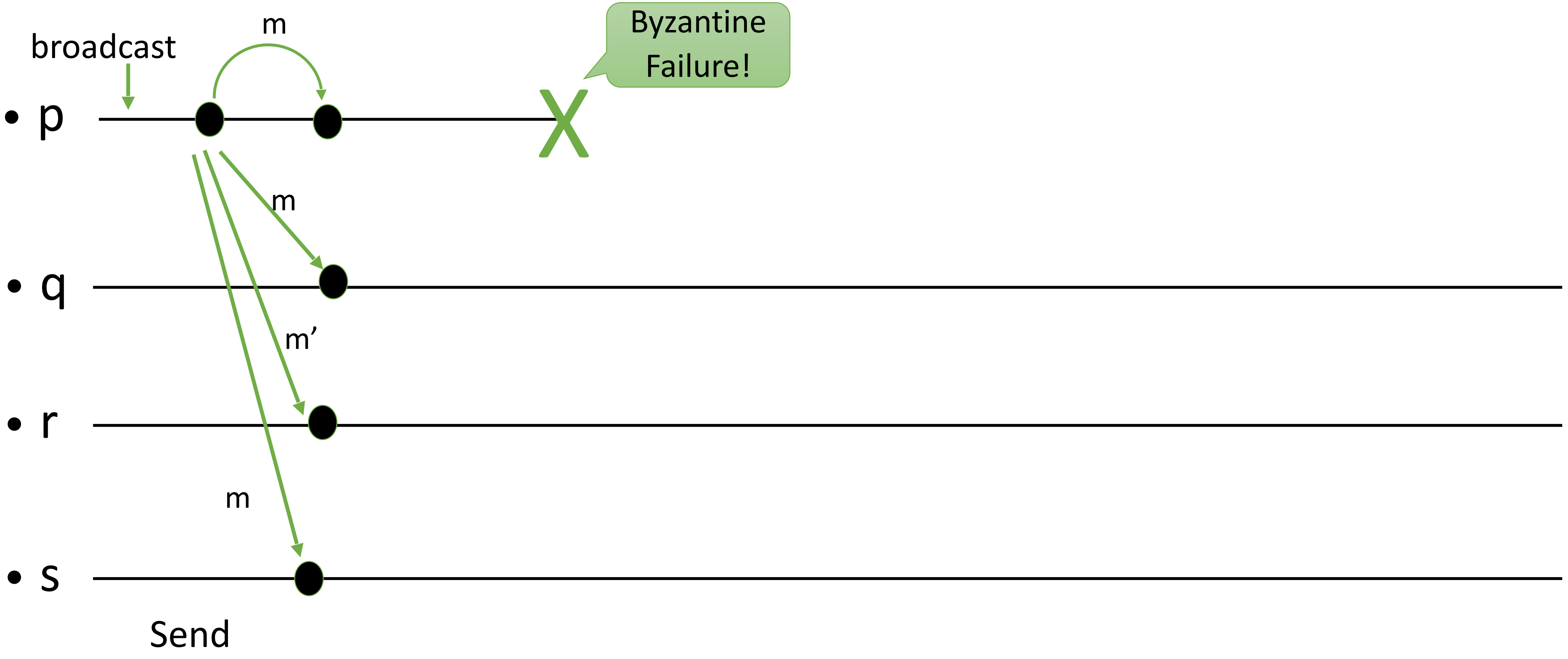
Example, Byzantine sender



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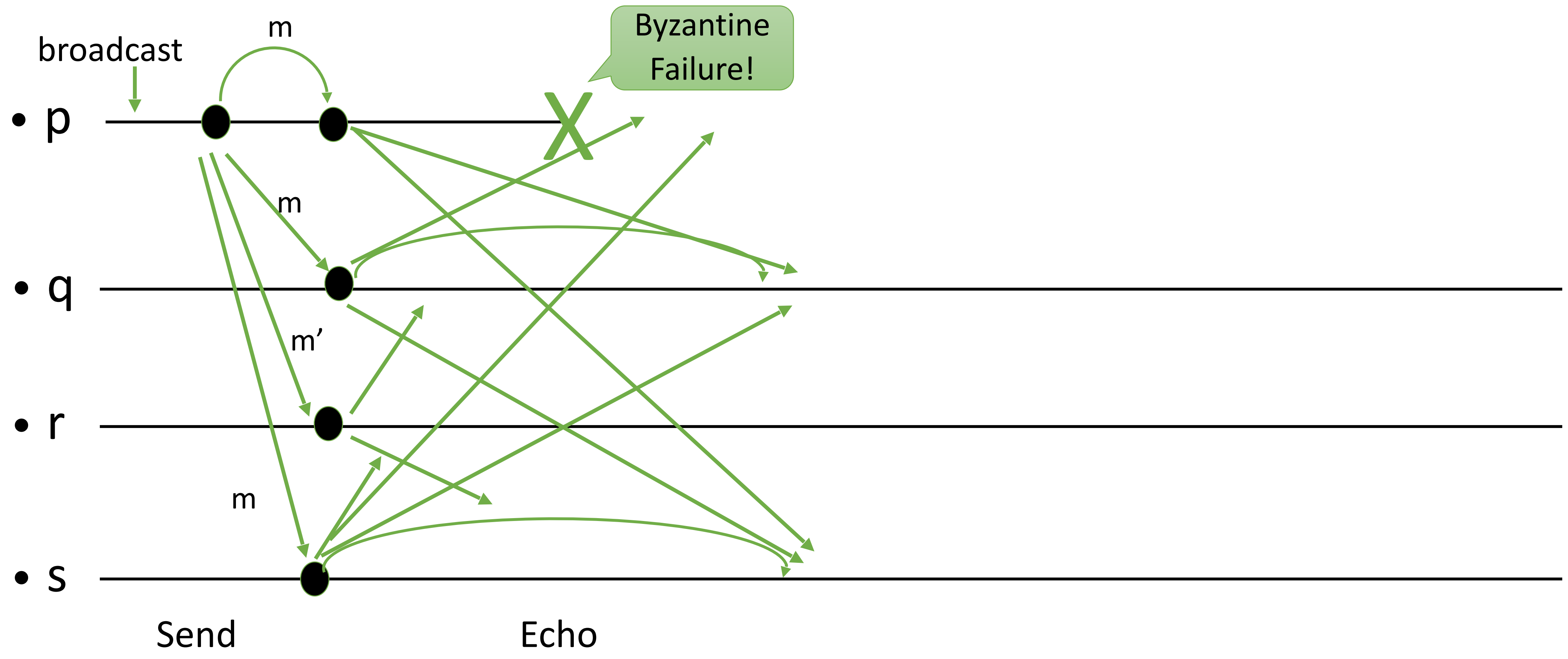

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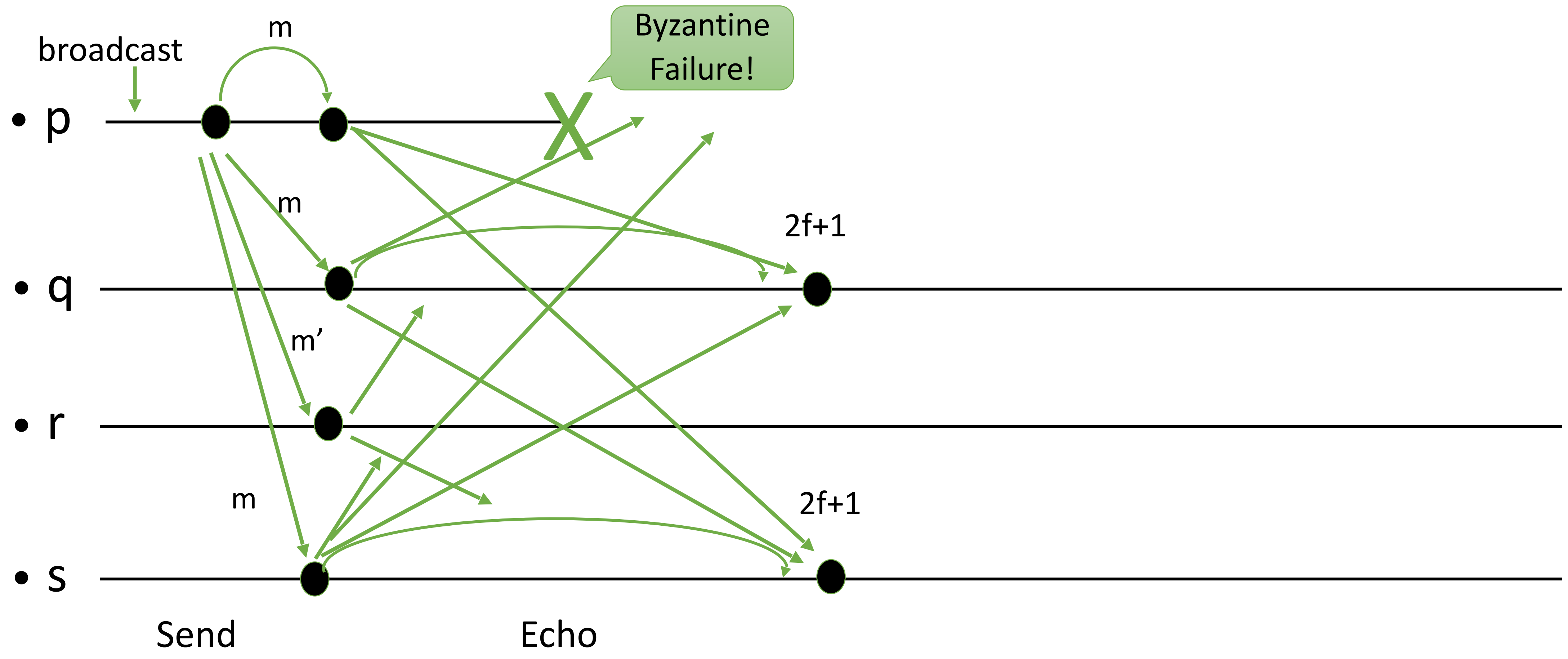

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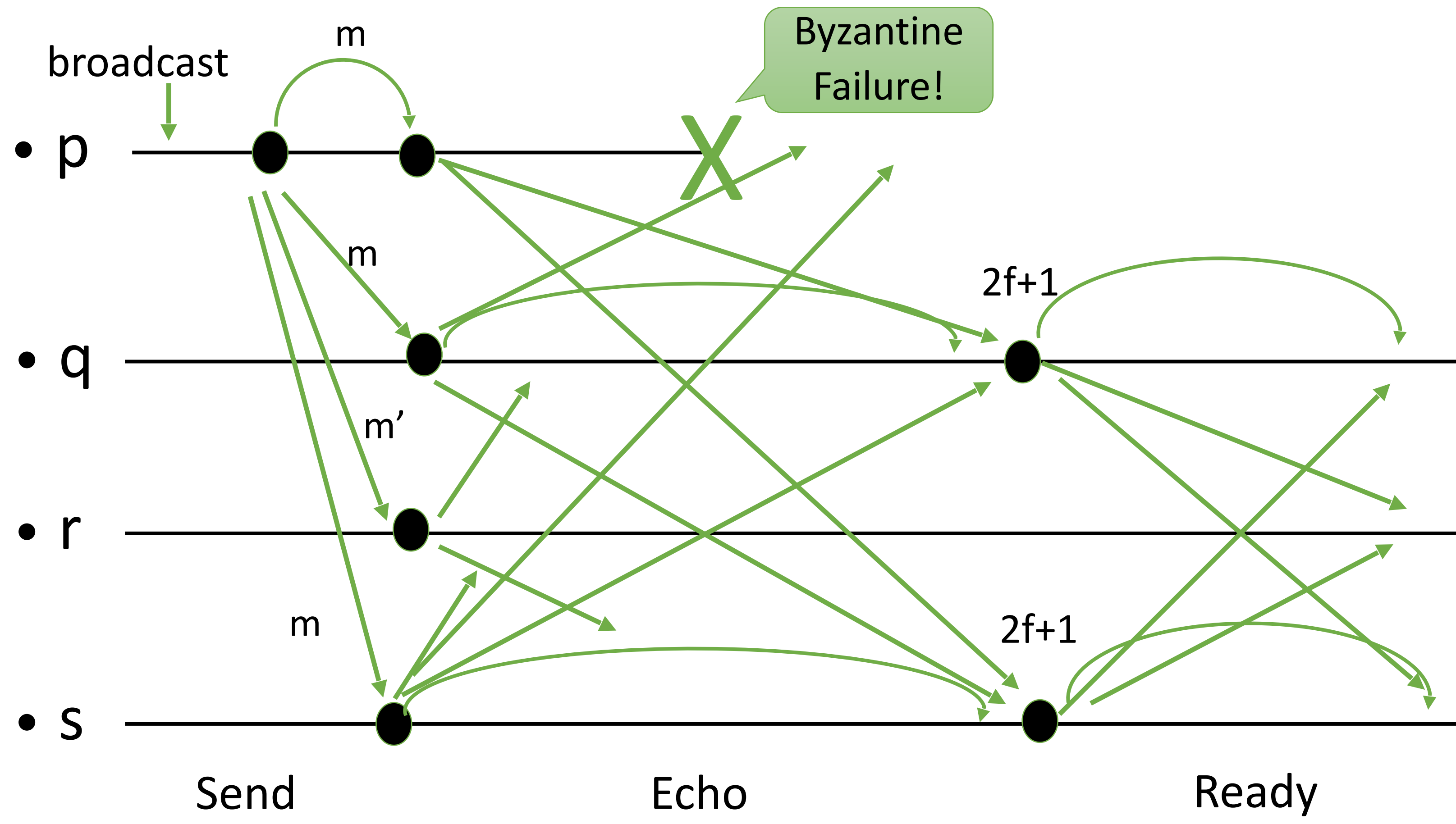

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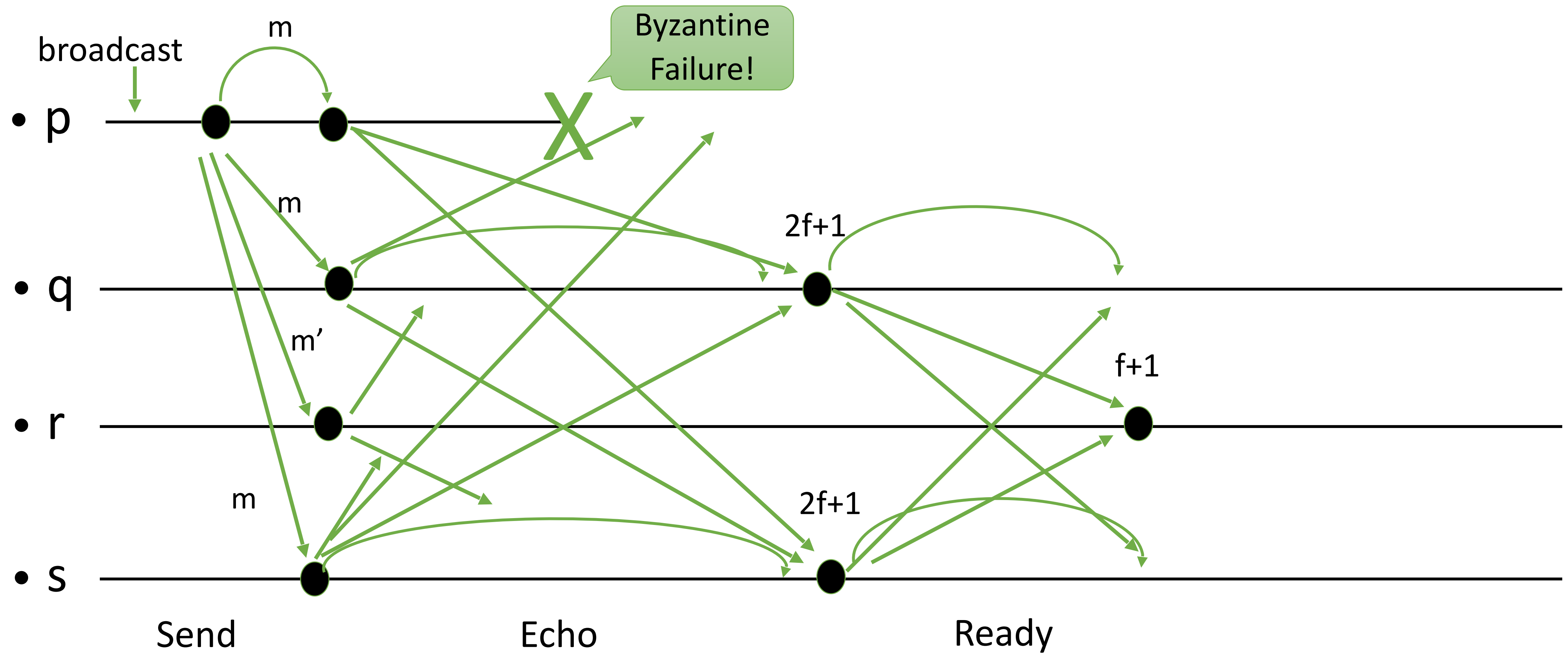

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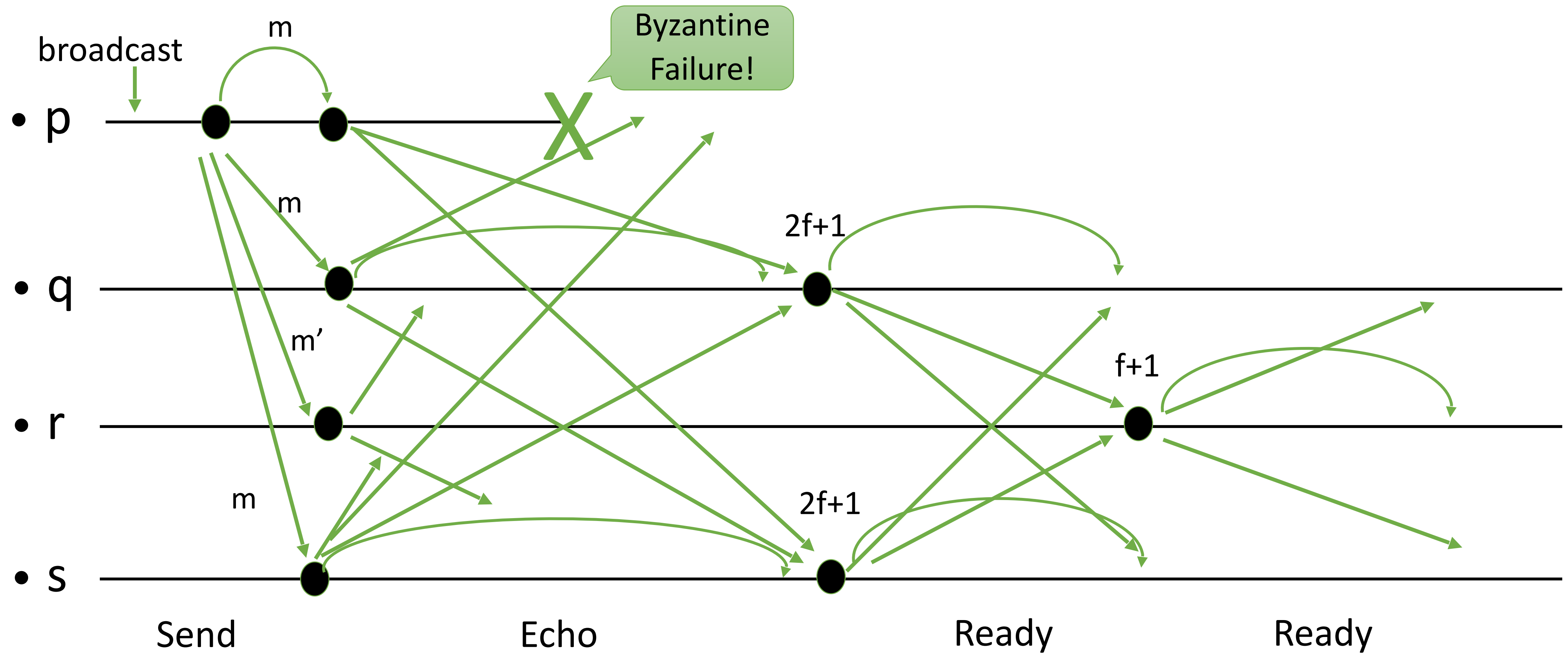

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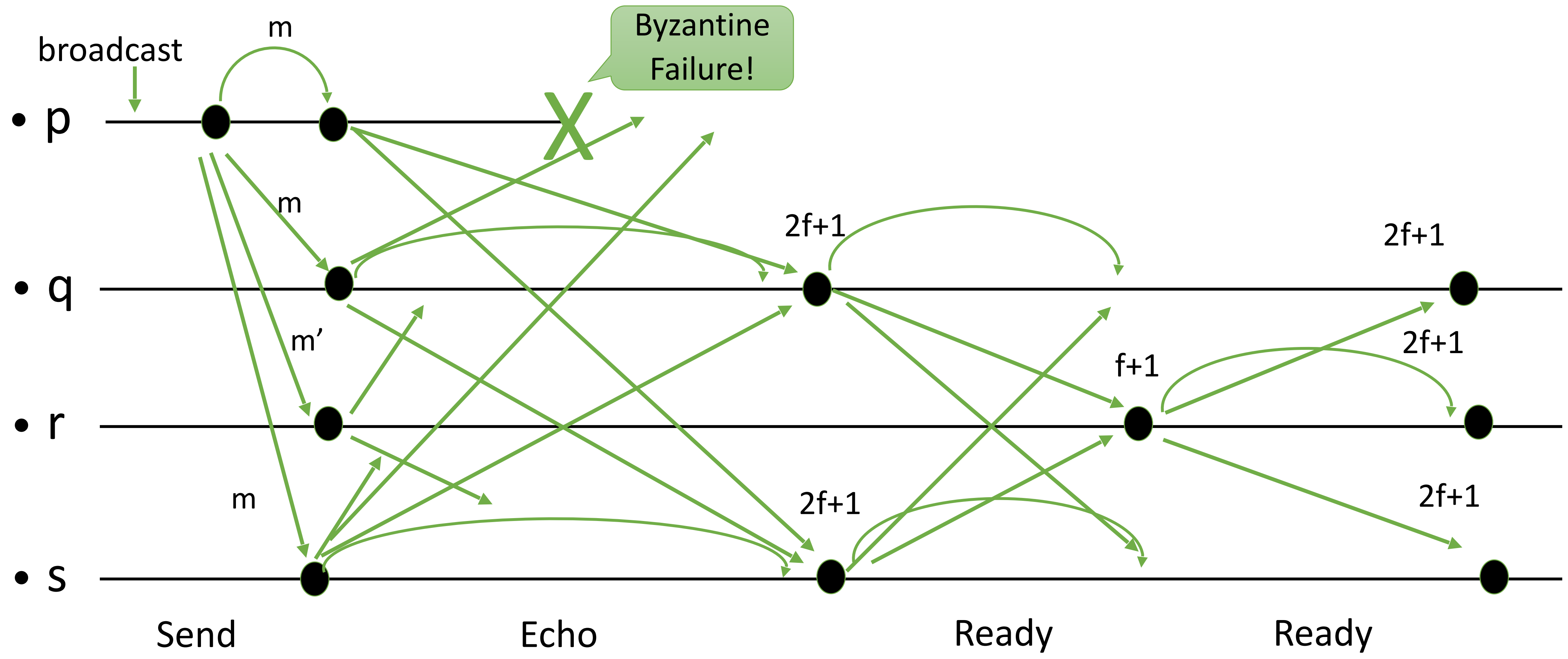

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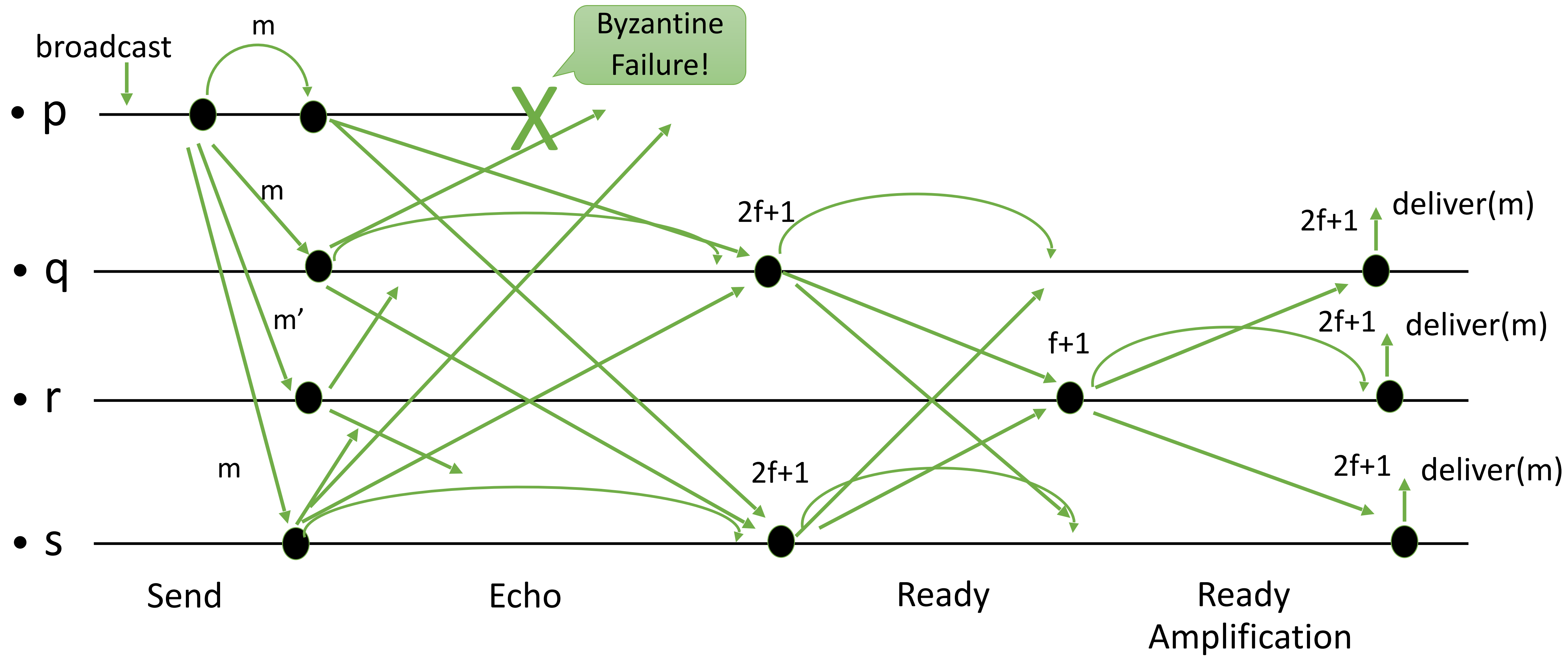

Authenticated Double-Echo Broadcast (Bracha)

Implements brb, **uses** abeb, with sender s (N>3f)

sent-ready := **false**

upon <brb, broadcast(m)> do

  trigger <abeb, broadcast(Send[m])>

upon <abeb, deliver(s, Send(m))> **do**

  trigger <abeb, broadcast(Echo[m])>

upon <abeb, deliver(p, Echo(m))> **do**

  echo[p] := m

  if $\exists m : \#\{p \mid echo[p]=m\} > 2f \land \neg$sent-ready **then**

    sent-ready := **true**

    trigger <abeb, broadcast(Ready(m))>

Authenticated Double-Echo Broadcast (Bracha)

upon <abeb, deliver(p, Ready(m))> **do**

  ready[p] := m

  if ∃m : #{p | ready[p]=m} > f ∧ ¬sent-ready **then**

    ▷ amplification of Ready messages

    sent-ready := **true**

    trigger <abeb, broadcast(Ready(m))>

  else if ∃m : #{p | ready[p]=m} > 2f **then**

    trigger <deliver(s, m)>

Note: some code to prevent duplicate execution is omitted.

Amplification

Proof of the totality property by Amplification:

- A correct process has delivered the message. In the 2f+1 processes that it received Ready from, there are at least f+1 correct processes. These correct processes send Ready to all processes.
- After receiving f+1 Ready messages, each correct processes amplifies the Ready message if it has not already sent it.
- There are at least 2f+1 correct processes, and they all send the Ready message.
- Therefore, each correct process receives Ready from 2f+1 processes and delivers the message.

Byzantine Broadcast Channel

- Implemented by a sequence of one-message instances of Byzantine broadcasts for each process
- Every message is delivered together with a unique label
  - Consistency and totality hold for each label
- Two variants
  - Consistent Channel
  - Reliable Channel

Interface and properties of Byzantine consistent channel

Module:

Name: ByzantineConsistentBroadcastChannel, instance bcch.

We introduce labels to uniquely identify deliveries.

Events:

- **Request**: 〈 bcch , broadcast (m) 〉: Broadcasts a message m to all processes.
- **Indication**: 〈 bcch , deliver(p, ℓ, m)〉: Delivers a message m with label ℓ broadcast by process p.

Properties:

- **BCCH1**: Validity: If a correct process p broadcasts a message m, then every correct process eventually delivers m.
- **BCCH2**: No duplication: For every process p and label ℓ, every correct process delivers at most *one message with label ℓ* and sender p.
- **BCCH3**: Integrity: If some correct process delivers a message m with sender p, and process p is correct, then m was previously broadcast by p.
- **BCCH4**: Consistency: If some correct process delivers a message *m with label ℓ* and sender s, and another correct process delivers a message *m′ with label ℓ* and sender s, then m = m′.

Byzantine Consistent Channel

Implements:

ByzantineConsistentBroadcastChannel, instance bcch.

Uses:

ByzantineConsistentBroadcast (multiple instances).

upon event ⟨ init ⟩ **do**

n := $[0]^N$

ready := **true**

forall p ∈ Π do

Initialize a new instance bcb[p, 0] of

ByzantineConsistentBroadcast with sender p

n: It keeps the number of the current broadcast instance for each process.
ready: It is used to wait for the previous broadcast instance to finish.

Byzantine Consistent Channel

upon event ⟨ broadcast(m) ⟩ **such that** ready = **true do**

  trigger ⟨ bcb[**self**, n[**self**]], broadcast(m) ⟩

  ready := **false**

upon event ⟨ bcb[p, n[p]], deliver(p, m) ⟩ **do**

  trigger ⟨ deliver(p, n[p], m) ⟩

  n[p] := n[p] + 1

  Initialize a new instance bcb[p, n[p]] of ByzantineConsistentBroadcast with sender p

  if p = self **then**

    ready := **true**